14.2.7 Denaturation of proteins

Each protein has a unique three-dimensional structure formed by interactions such as disulphide bond, hydrogen bond, hydrophobic and electrostatic interactions. These interactions can be disturbed when the protein is exposed to a higher temperature, by adding certain chemicals such as urea, alteration of pH and ionic strength etc., It leads to the loss of the three-dimensional structure partially or completely. The process of a losing its higher order structure without losing the primary structure, is called denaturation. When a protein denatures, its biological function is also lost.

Since the primary structure is intact, this process can be reversed in certain proteins. This can happen spontaneously upon restoring the original conditions or with the help of special enzymes called cheperons (proteins that help proteins to fold correctly).

Example: coagulation of egg white by action of heat.

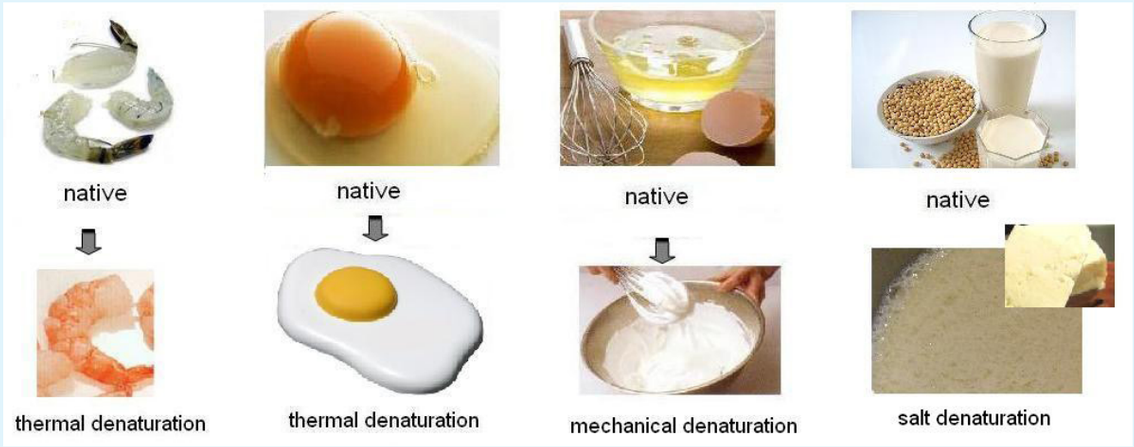


Figure 14.18 Denaturation of proteins

14.2.8 Importance of proteins

Proteins are the functional units of living things and play vital role in all biological processes

1. All biochemical reactions occur in the living systems are catalysed by the catalytic proteins called enzymes.
2. Proteins such as keratin, collagen act as structural back bones.
3. Proteins are used for transporting molecules (Haemoglobin), organelles (Kinesins) in the cell and control the movement of molecules in and out of the cells (Transporters).
4. Antibodies help the body to fight various diseases.
5. Proteins are used as messengers to coordinate many functions. Insulin and glucagon control the glucose level in the blood.
6. Proteins act as receptors that detect presence of certain signal molecules and activate the proper response.
7. Proteins are also used to store metals such as iron (Ferritin) etc.

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